DataSquad: An entry point for RSE

Student RSE model for research support, experiential learning, and technical workforce development

Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

Paid students contribute to research and build RSE skills while they do it.

Campuses need more data and software help than staff can provide. Courses leave gaps, and so does the code researchers write. DataSquads fill those gaps.

Research requests → **Student project work** (Staff teach, review, and mentor) → **Useful research support** + **Skilled students**



Experience



Mentorship



Professional Growth



Applied Curricular Skills

What the work looks like

CARLETON Election-news scraper. Seven students built it in 2022. Two more extended it from 2023 to 2024 for a political science project that involved over 500,000 transcripts.



Each dot is one student. Blue built it, orange extended it.

UCLA Police helicopter flight data. Two students brought software skills to a research lab's R code. They refactored it with staff, and the code went through code review. Loading the data fell from about 318 minutes to about 2.



LAPD helicopter, for illustration. Photo: Chris Yarzab, CC BY 2.0, via Wikimedia Commons. Cropped and resized.

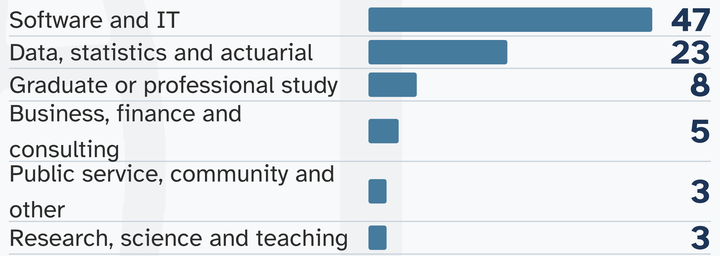
Time to load the data

Before	After
318 min	2 min

Research software habits, when the project needs them. Run instructions, a dependency list, one test, and a citation file. We do not treat them as a checklist. We plan to track whether researchers keep them. Two student-built projects have citation files and DOIs that credit their developers: wbcheck (**doi:10.5281/zenodo.23329531**) and the UC OSPO Education site (**doi:10.5281/zenodo.23330022**).

Where students go next

Current job category of former student employees, from public profiles in 2026.



89 of 132 alumni coded so far: UCLA 19 of 19, Carleton 70 of 113, 28 of them coded from the job title alone. Carleton is only partly coded, so this is not a sample. Descriptive, not a placement rate.

What adopting takes

- Student positions and a wage budget
- Supervisors with time to review and coach
- Onboarding and a home for documentation
- A plan for turnover

Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. Supervise students? Compare onboarding, review, and graduation handoffs with us.

datasquad.info/join